

Atividade - Aula 003

Derivação de um Código a partir da Gramática de uma Linguagem de Programação

1. Linguagem escolhida:

A linguagem escolhida foi: JavaScript;

2. Fonte:

Seção 5.1 — Syntactic and Lexical Grammars da fonte:

https://262.ecma-international.org/17.0/index.html?_gl=1*ngtd40*_ga*MTM5NzY4Njk2LjE3ODc3OTA3NDQ.*_ga_TDCK4DWEPP*czE3ODc3OTA3NDMkbzEkZzAkDE3ODc3OTA3NDMkajYwJGwwJGgw#sec-syntactic-and-lexical-grammars

3. Código que será derivado:

```
var isTested = false;

if (isTested) {
    console.log("It was tested");
}
```

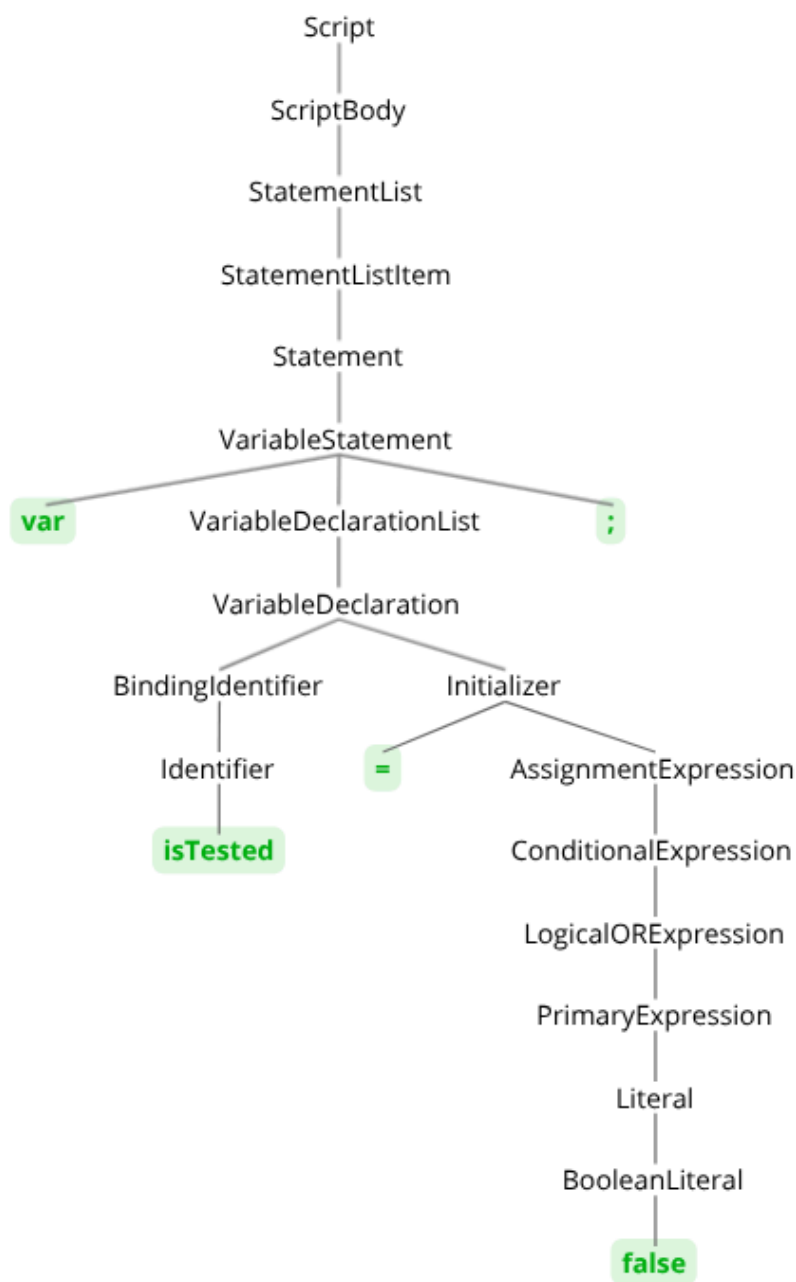
4. Regras de produção selecionadas:

VariableStatement : var VariableDeclarationList ;
PrimaryExpression : Identifier | Literal
ScriptBody : StatementList
Block : { StatementList }
Arguments : (ArgumentList)
ExpressionStatement : Expression ;
Literal : BooleanLiteral | StringLiteral
LogicalORExpression : PrimaryExpression
Initializer : = AssignmentExpression
BindingIdentifier : Identifier
IdentifierName : Identifier | log
BooleanLiteral : false
VariableDeclarationList : VariableDeclaration
StatementList : StatementListItem | StatementList StatementListItem
Script : ScriptBody
Expression : AssignmentExpression
AssignmentExpression : ConditionalExpression | CallExpression
ConditionalExpression : LogicalORExpression
MemberExpression : PrimaryExpression | MemberExpression .
IdentifierName
StatementListItem : Statement

ArgumentList : AssignmentExpression
 IfStatement : if (Expression) Statement
 StringLiteral : "It was tested"
 Punctuators : =, ,, (,), {, }, .
 CallExpression : MemberExpression Arguments
 Keywords : var, if
 VariableDeclaration : BindingIdentifier Initializer
 Statement : VariableStatement | IfStatement | Block |
 ExpressionStatement
 Identifier : isTested, console

5. Derivação: Os terminais estão identificados em verde.

a. `var isTested = false;`



```

b. if (isTested) {
    console.log("It was tested");
}

```

